

Aim: Learn LED control and button input on Arduino.

In today's lab, you will control the "LED"s of Ardukit, performing the tasks below with your project members. Work in pairs of **two**.

1. As you examine the kit, you can see the pin numbers for each LED and button. As your first task, define digital pins for four LEDs and two buttons.
2. In your setup method, initialize all LED pins as output and all button pins as input.
3. In the loop method, turn LED1 on when Button1 is pressed. When Button1 is released, LED1 should turn off.
4. Modify your code so that Button1 toggles LED1 whenever it is clicked. That means the first click turns LED1 on, the second click turns it off, and the next click turns it on again.
5. Modify your code so that Button1 works as an ON/OFF switch for the whole LED system. When the system is OFF, all LEDs should remain turned off.
6. Define a second button. Each time Button2 is clicked, the system should switch to the next LED mode. The mode order should be:

Mode 1 -> Mode 2 -> Mode 3 -> Mode 1 -> ...

7. Implement the following three LED modes. These modes should only run when the system is ON:
 - a. **Mode 1:** All LEDs blink together with 1-second intervals.
 - b. **Mode 2:** LEDs turn on sequentially from left to right with 1-second intervals.
 - c. **Mode 3:** LEDs turn on sequentially from right to left with 1-second intervals.

HINT: For button toggle and mode switching, you should save the previous button states for the next loop. Otherwise, one button press may be detected multiple times.

As a starting point, you can use the following code snippet:

```
int LED1pin = 43;  
int LED2pin = 44;  
int LED3pin = 45;  
int LED4pin = 46;
```

```
int button1pin = 47;  
int button2pin = 48;
```

```
void setup() {  
  pinMode(LED1pin, OUTPUT);  
  pinMode(LED2pin, OUTPUT);  
  pinMode(LED3pin, OUTPUT);  
  pinMode(LED4pin, OUTPUT);
```

```
  pinMode(button1pin, INPUT);  
  pinMode(button2pin, INPUT);  
}
```

```
void loop() {  
  // Read button states here.
```

```
  // Write your LED control code here.  
}
```